Maximum Likelihood Estimation

Theory Questions

TTK4260

Question 1 *(Probability vs Likelihood)*

Consider a Normal distribution with unknown mean μ and known variance $\sigma^2 = 1$. You observe a single data point $x = 2.3$.

(a) Explain the key conceptual difference between *probability* and *likelihood*. What is fixed and what varies in each case? *[4 points]*

(b) Write down the likelihood function $L(\mu \mid x = 2.3)$ for this observation. *[3 points]*

(c) Does the likelihood function integrate to 1 over all possible values of μ? Why or why not? *[3 points]*

Question 2 *(The Maximum Likelihood Principle)*

(a) State the Maximum Likelihood Principle in words and write the formal definition of the MLE $\hat{\theta}_{\mathrm{ML}}$. *[4 points]*

(b) Compare Maximum Likelihood with Least Squares. List two things that ML provides that LS does not. *[4 points]*

(c) What is the main caveat or disadvantage of using Maximum Likelihood compared to Least Squares? *[2 points]*

Question 3 *(Why Log-Likelihood?)*

For i.i.d. observations $x_1, x_2, \ldots, x_n$, the likelihood function is $L(\theta) = \prod_{i=1}^{n} f(x_i \mid \theta)$.

(a) Explain why computing $L(\theta)$ directly can cause numerical problems when n is large. Give a concrete numerical example. *[4 points]*

(b) Define the log-likelihood $\ell(\theta)$ and explain why $\arg\max_\theta L(\theta) = \arg\max_\theta \ell(\theta)$. *[3 points]*

(c) List two additional advantages of working with log-likelihood beyond numerical stability. *[3 points]*

Question 4 *(The Score Function)*

The score function is defined as $s(\theta) = \frac{\partial}{\partial\theta}\ell(\theta)$.

(a) What equation must be satisfied at the MLE? State this in terms of the score function. *[2 points]*

(b) What second-order condition should you check to confirm that a solution to the score equation is actually a maximum (not a minimum)? *[3 points]*

(c) For the Poisson distribution with PMF $P(X = k \mid \lambda) = \frac{\lambda^k e^{-\lambda}}{k!}$, write down the log-likelihood for a single observation x. *[3 points]*

Question 5 *(MLE for Normal Distribution: Derivation)*

Let $X_1, X_2, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with both parameters unknown.

(a) Write down the likelihood function $L(\mu, \sigma^2)$ for these observations. *[3 points]*

(b) Derive the log-likelihood $\ell(\mu, \sigma^2)$ and simplify it. *[4 points]*

(c) Find $\hat{\mu}_{\text{ML}}$ by taking the derivative of ℓ with respect to μ, setting it to zero, and solving. *[4 points]*

(d) Find $\hat{\sigma}^2_{\text{ML}}$ by taking the derivative of ℓ with respect to σ^2, setting it to zero, and solving. *[4 points]*

Question 6 *(MLE for Bernoulli Distribution)*

A coin is flipped n times, yielding outcomes $X_1, X_2, \ldots, X_n$ where $X_i \in \{0, 1\}$ and $P(X_i = 1) = p$.

(a) Write down the likelihood function $L(p)$ for these observations. Simplify it in terms of $\sum_{i=1}^{n} x_i$. *[4 points]*

(b) Derive the MLE $\hat{p}_{\text{ML}}$ by maximizing the log-likelihood. Show your work. *[5 points]*

Question 7 *(MLE for Poisson Distribution)*

A call center records the number of calls received per hour. The data follows a Poisson distribution with unknown rate λ.

(a) Write down the likelihood function for n observations $x_1, \ldots, x_n$. *[3 points]*

(b) Derive the log-likelihood and find the MLE $\hat{\lambda}_{\text{ML}}$. *[5 points]*

(c) Given data: 4, 7, 5, 6, 8, 3, 5, 6, 7, 4 calls per hour over 10 hours, compute $\hat{\lambda}_{\text{ML}}$. *[2 points]*

(d) For the Poisson distribution, what is the relationship between the mean and variance? How can you use this to check whether Poisson is an appropriate model? *[3 points]*

Question 8 *(MLE for Exponential Distribution)*

Component lifetimes follow an Exponential distribution with rate parameter λ, i.e., $f(x \mid \lambda) = \lambda e^{-\lambda x}$ for $x \geq 0$.

(a) Write down the likelihood function for n observed lifetimes $x_1, \ldots, x_n$. *[3 points]*

(b) Derive the MLE $\hat{\lambda}_{\text{ML}}$. *[5 points]*

Question 9 *(ML = LS Connection)*

Consider the linear regression model $y_i = \boldsymbol{x}_i^\top \boldsymbol{\theta} + \varepsilon_i$ where $\varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$.

(a) Write down the likelihood function $L(\boldsymbol{\theta}, \sigma^2 \mid \boldsymbol{y}, \boldsymbol{X})$ for this model. *[4 points]*

(b) Show that maximizing the log-likelihood with respect to $\boldsymbol{\theta}$ is equivalent to minimizing $\sum_{i=1}^{n} (y_i - \boldsymbol{x}_i^\top \boldsymbol{\theta})^2$. *[5 points]*

(c) Under what assumption about the error distribution does ML = LS? *[2 points]*

Question 10 *(Choosing the Right Distribution)*

(a) For each scenario, name the most appropriate distribution:

- Number of customer arrivals per hour at a store
- Whether a manufactured part passes or fails quality inspection
- Time until a light bulb burns out
- Heights of adult males in a population

[4 points]

(b) Explain why using a Normal distribution for count data (e.g., number of defects) would be problematic. *[3 points]*

(c) You have count data where the sample variance is much larger than the sample mean. What does this suggest, and what distribution might be more appropriate than Poisson? *[3 points]*